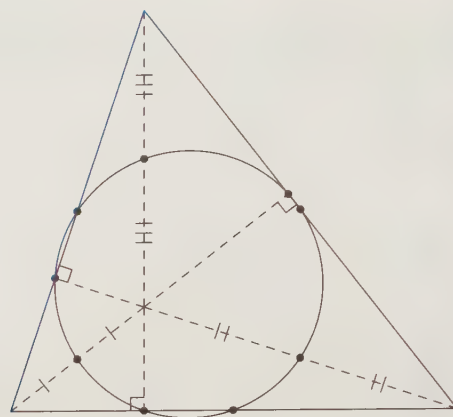




The Nine-Point Circle

Why isn't there a special name for the tops of your feet? – Lily Tomlin

CHAPTER 7

Special Parts of a Triangle

Most of mastering geometry is mastering triangles. In this chapter we explore several special points, lines, and circles that can be found in any triangle.

In general, any line segment from a vertex of a triangle to a point on the opposite side is called a **cevian**. Before we dive into the special parts of a triangle, we'll investigate a couple of special lines.

7.1 Bisectors

The **perpendicular bisector** of a line segment is the line passing through the midpoint of the segment such that the line is perpendicular to the segment. In Figure 7.1, line k is the perpendicular bisector of $\overline{AB}$.

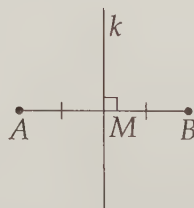


Figure 7.1: A Perpendicular Bisector

The **angle bisector** of an angle is the ray that divides the angle into two equal angles. In Figure 7.2, ray m is the angle bisector of $\angle AOB$.

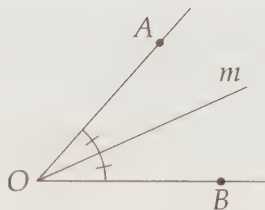
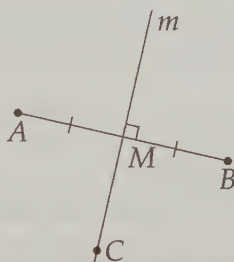


Figure 7.2: An Angle Bisector

Problems

Problem 7.1: Line m is the perpendicular bisector of $\overline{AB}$. Use congruent triangles to show that if C is on m , then $CA = CB$.



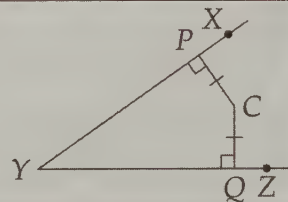
Problem 7.2: In this problem, we will show that if $AC = BC$, then C is on the perpendicular bisector of $\overline{AB}$.

- Let C be a point such that $AC = BC$ and let M be the midpoint of $\overline{AB}$. Draw a diagram.
- Draw $\overline{CM}$, then use congruent triangles to prove that $\overline{CM} \perp \overline{AB}$. Why does this mean that C is on the perpendicular bisector of $\overline{AB}$?

Problem 7.3: In this problem we show that any point on the angle bisector of $\angle XYZ$ is equidistant from $\overrightarrow{YX}$ and $\overrightarrow{YZ}$.

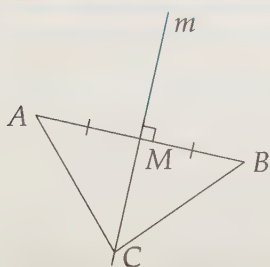
- Let C be on the angle bisector of $\angle XYZ$. Draw $\angle XYZ$ and segments from C to the sides of $\angle XYZ$ that are perpendicular to $\overrightarrow{YX}$ and $\overrightarrow{YZ}$, respectively.
- Use congruent triangles to prove that the two segments from C to the sides of the angle are equal in length.

Problem 7.4: C is a point inside $\angle XYZ$ such that C is equidistant from $\overrightarrow{YZ}$ and $\overrightarrow{YX}$ as shown. Show that C must be on the angle bisector of $\angle XYZ$.



Before we start tackling problems involving special lines in triangles, we will use what we already know about congruent triangles to learn about these special lines. We'll start with perpendicular bisectors.

Problem 7.1: Line m is the perpendicular bisector of $\overline{AB}$. Show that if C is on m , then $CA = CB$.

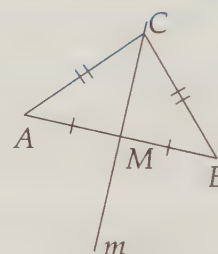


Solution for Problem 7.1: Drawing $\overline{AC}$ and $\overline{CB}$, we see that $\triangle AMC \cong \triangle BMC$ by SAS Congruence since $AM = MB$, $\angle AMC = \angle BMC$, and $CM = CM$. Therefore, $AC = CB$. $\square$

Having shown that every point on the perpendicular bisector of a segment is equidistant from the endpoints of the segment, we should wonder if every point that is equidistant from these endpoints has to be on the perpendicular bisector of the segment connecting them.

Problem 7.2: Show that if $AC = BC$, then C is on the perpendicular bisector of $\overline{AB}$.

Solution for Problem 7.2: To show that C is on the perpendicular bisector of $\overline{AB}$, we connect C to the midpoint of $\overline{AB}$, which we'll call M , then show that $\overline{CM} \perp \overline{AB}$. Since M is the midpoint of $\overline{AB}$, we have $AM = MB$. We are given $AC = CB$, and obviously $CM = CM$, so we have $\triangle ACM \cong \triangle BCM$ by SSS Congruence. Therefore, $\angle AMC = \angle BMC$. Since these two angles must add to 180° , they must each equal 90° . So, $\overline{CM}$ is the perpendicular bisector of $\overline{AB}$. (This proof doesn't address the possibility that C is the midpoint of $\overline{AB}$. We take care of this by noting that the midpoint of $\overline{AB}$ is on the perpendicular bisector of $\overline{AB}$ by definition.) $\square$



Putting these last two problems together tells us something very important about the perpendicular bisector of a segment.

Important: The perpendicular bisector of a segment is the straight line consisting of all points that are equidistant from the endpoints of the segment.



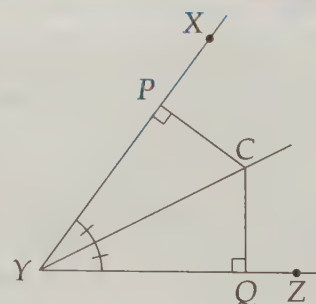
We can prove something very similar for angle bisectors.

Problem 7.3: Show that any point on the angle bisector of $\angle XYZ$ is equidistant from $\overrightarrow{YX}$ and $\overrightarrow{YZ}$.

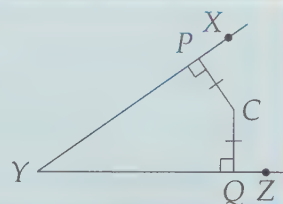
Solution for Problem 7.3: We start by drawing the lengths we need to show equal. These are the perpendicular segments $\overline{CP}$ and $\overline{CQ}$ from C to $\overrightarrow{YX}$ and $\overrightarrow{YZ}$ as shown. By AAS Congruence, we have $\triangle CPY \cong \triangle CQY$, so $CP = CQ$.

Therefore, any point on the angle bisector of an angle is equidistant from the sides of the angle. $\square$

As with the perpendicular bisector, we can 'run this backwards,' deducing that any point that is equidistant from the sides of an angle must be on the angle bisector.

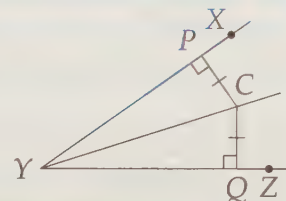


Problem 7.4: C is a point inside $\angle XYZ$ such that C is equidistant from $\overrightarrow{YZ}$ and $\overrightarrow{YX}$ as shown. Show that C must be on the angle bisector of $\angle XYZ$.



Solution for Problem 7.4: We have $\triangle CPY \cong \triangle CQY$ by HL Congruence because $CP = CQ$ and $CY = CY$. Therefore, $\angle CYP = \angle CYQ$, so $\overrightarrow{YC}$ bisects $\angle XYZ$. $\square$

Putting the last two problems together tells us a little more about angle bisectors.



Important: The angle bisector of an angle consists of all points that are equidistant from the sides of the angle.



Exercises

7.1.1 Line m is the perpendicular bisector of both $\overline{AB}$ and $\overline{CD}$. Which of the following must be true and why (or why not)?

- (a) $AB = CD$.
- (b) $\overline{AB} \parallel \overline{CD}$.
- (c)★ $AC = BD$. **Hints:** 460

7.1.2 Describe all the points that are equidistant from two intersecting lines.

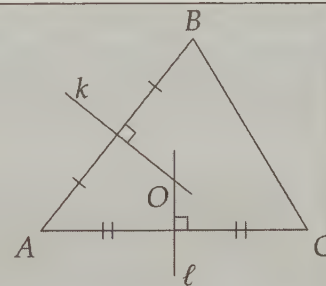
7.1.3 Describe all the points that are equidistant from two parallel lines.

7.2 Perpendicular Bisectors of a Triangle

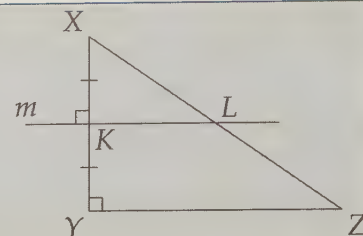
Problems

Problem 7.5: Let k and ℓ be perpendicular bisectors of the sides $\overline{AB}$ and $\overline{AC}$ of $\triangle ABC$ as shown.

- (a) From what two points must every point on k be equidistant?
- (b) From what two points must every point on ℓ be equidistant?
- (c) Let O be the intersection of k and ℓ . Why must O be on the perpendicular bisector of $\overline{BC}$?
- (d) If we draw a circle with center O and radius OA , will the circle go through B and C as well? Why or why not?



Problem 7.6: Line m is the perpendicular bisector of side $\overline{XY}$ of right triangle $\triangle XYZ$, and m meets $\overline{XZ}$ at L as shown.



- Show that $\triangle XKL \sim \triangle XYZ$.
- Use part (a) to show that L is the midpoint of $\overline{XZ}$.
- Where do the perpendicular bisectors of the sides of a right triangle meet?
- What is the radius of the circle that passes through X , Y , and Z ?

Problem 7.7: Ariel and Zappa are playing 'NAME THAT CIRCLE.' In 'NAME THAT CIRCLE,' Ariel imagines a circle drawn on her sheet of paper. Zappa can then ask for points that are on the circle. Zappa wins when he can draw the whole circle.

- Can Zappa win the game if Ariel only gives him two points?
- Can Zappa win if Ariel gives him three points?
- Zappa asks for four points. Ariel isn't really paying attention and just picks out four points at random on the piece of paper and ends up with points shown in the diagram at right. Zappa tries and tries, but he can't NAME THAT CIRCLE. Finally, he insists that there is no such circle. Could he possibly be right or should he just try harder?

Ariel's points

Problem 7.8: In this problem we find the radius of a circle that passes through all three vertices of an equilateral triangle. Let $\triangle ABC$ be our triangle, with $AB = 6$. Let P be the midpoint of $\overline{BC}$ and Q the midpoint of $\overline{AC}$, and let O be the center of the circle through the vertices of $\triangle ABC$.

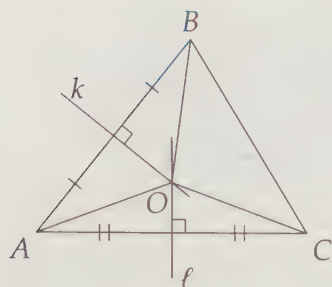
- Draw a diagram. Include the perpendicular bisectors of $\overline{AC}$ and $\overline{BC}$. Where must these perpendicular bisectors intersect?
- Prove that $\triangle OCP$ is a 30-60-90 right triangle.
- Find CP , then find the radius of the circle.

Now that we've learned a little bit about perpendicular bisectors, we're ready to study the perpendicular bisectors of the sides of a triangle.

Problem 7.5: Show that the three perpendicular bisectors of the sides of a triangle are **concurrent**, meaning they meet at a single point. Show that there is a circle centered at this point that passes through all three vertices of the triangle.

Extra! *The mathematician's patterns, like those of the painter or the poet, must be beautiful; the ideas, like the colors or the words, must fit together in a harmonious way. There is no permanent place in the world for ugly mathematics.*

—G. H. Hardy



Solution for Problem 7.5: We start by thinking about just two of the perpendicular bisectors, k and ℓ , shown in the figure at left. Let k and ℓ meet at O . We wish to show that O is on the perpendicular bisector of $\overline{BC}$.

Since O is on the perpendicular bisector of $\overline{AB}$, it is equidistant from A and B , i.e. $OA = OB$. Similarly, since O is on the perpendicular bisector of $\overline{AC}$, we have $OA = OC$. So, $OA = OB = OC$. Since $OB = OC$, O must be on the perpendicular bisector of $\overline{BC}$ (see Problem 7.2), so the perpendicular bisectors of the sides of a triangle are concurrent.

Seeing $OA = OB = OC$, we note that if we draw a circle with center O and radius OA , the circle will pass through B and C as well. $\square$

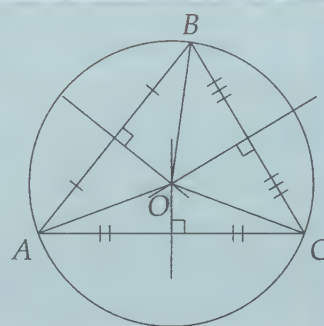
The point O and the circle described in the last problem are so useful in geometry problems that they have their own names.

Important:



The perpendicular bisectors of the sides of a triangle are concurrent at a point called the **circumcenter**. The circle centered at the circumcenter that passes through the vertices of the original triangle is called the **circumcircle** of the triangle because it is **circumscribed** about the triangle (meaning it passes through all the vertices of the triangle).

Finally, the radius of this circle is called the **circumradius**, the circumcenter is usually labeled with the letter O , and the circumradius is usually called R .



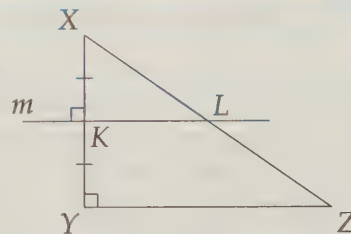
In right triangles, the circumcircle is particularly easy to describe.

Problem 7.6: Where is the circumcenter of a right triangle? What is the circumradius of a right triangle?

Solution for Problem 7.6: If we draw a few right triangles and their perpendicular bisectors, we will begin to suspect that the midpoint of the hypotenuse is the circumcenter.

We start our proof by drawing the perpendicular bisector of one of the legs, such as line m in the diagram. Let m meet $\overline{XZ}$ at L . Since m and $\overline{YZ}$ are both perpendicular to $\overline{XY}$, we have $m \parallel \overline{YZ}$. Therefore, $\triangle XKL \sim \triangle XYZ$, so $XL/XZ = XK/XY$. Since $XK/XY = 1/2$, we have $XL/XZ = 1/2$, so L is the midpoint of $\overline{XZ}$. Clearly, L is also on the perpendicular bisector of $\overline{XZ}$ (since L is the midpoint of $\overline{XZ}$), so L must be where the perpendicular bisectors of $\triangle XYZ$ intersect. Therefore, L is the circumcenter of $\triangle XYZ$.

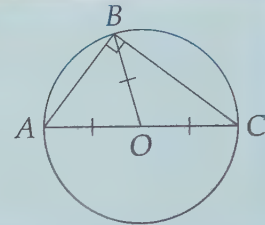
The endpoints of the hypotenuse are vertices of the triangle, so by definition they are on the circumcircle. Since the midpoint of the hypotenuse of our right triangle is the circumcenter, the hypotenuse is a diameter of the circumcircle. Therefore, the circumradius is half the length of the hypotenuse of the right triangle. $\square$



Important:



The circumcenter of a right triangle is the midpoint of the hypotenuse, and the circumradius equals one-half the length of the hypotenuse.



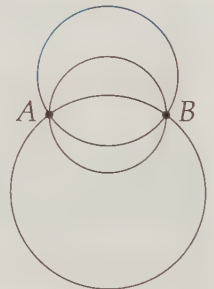
Back in Problem 1.1, we saw that two points are enough to define a line. All this talk about circles going through the vertices of a triangle should make us wonder – how many points of a circle do we need to define a circle? Clearly, one won't do, since we can draw all sorts of circles through a given point.

Problem 7.7: Ariel and Zappa are playing 'NAME THAT CIRCLE.' In 'NAME THAT CIRCLE,' Ariel imagines a circle drawn on her sheet of paper. Zappa can then ask for points that are on the circle. Zappa wins when he can draw the whole circle.

- Can Zappa win the game if Ariel only gives him two points?
- Can Zappa win if Ariel gives him three points?
- Zappa asks for four points. Ariel isn't really paying attention and just picks out four points at random on the piece of paper. Zappa tries and tries, but he can't NAME THAT CIRCLE. Finally, he insists that there is no such circle. Could he possibly be right or should he just try harder?

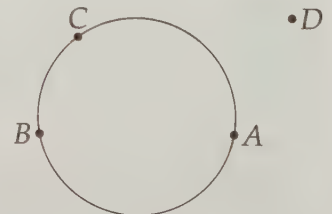
Solution for Problem 7.7:

- Clearly two points isn't enough – we can make all sorts of different circles given two points. This is shown at right.
- Given three points, Zappa can NAME THAT CIRCLE! He can connect the points to make a triangle. Then, he can draw the perpendicular bisectors of two of the sides of the triangle. Where they meet is the circumcenter. He can then make the circumcircle. Notice that the three points must be **noncollinear**, meaning that it's impossible to draw a single line that goes contains all three points.



The center of any circle that passes through all three points must be on all three perpendicular bisectors of the sides of the triangle with these points as vertices. Therefore, the circumcenter is the only possible center of a circle through the three points. Therefore, the circumcircle is the *only* circle that passes through all three points.

- Let the points be A , B , C , and D . As described in the previous part, there is only one circle through A , B , and C . But this circle doesn't have to go through D ! Therefore, if we pick out four points at random, there doesn't have to be a circle that passes through all four points. Zappa should give up once he sees that the circumcircle of $\triangle ABC$ doesn't pass through D . One such arrangement of points is shown at right.



□

Important:

Just as two points determine a line, we have now shown that three noncollinear points determine a circle. This means that given any three noncollinear points, there is exactly one circle that passes through all three.

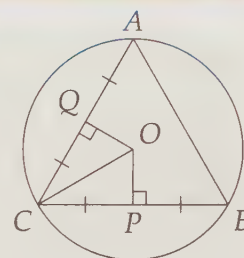
Now let's try a problem involving the circumradius.

Problem 7.8: Find the circumradius of an equilateral triangle with side length 6.

Solution for Problem 7.8: Let $\triangle ABC$ be our equilateral triangle. We connect its circumcenter to a vertex and draw a couple perpendicular bisectors as shown. Since $CQ = CP$ and $OC = OC$, we know that $\triangle OCP \cong \triangle OCQ$ by HL Congruence. Therefore, $\angle OCQ = \angle OCP$, so $\overline{OC}$ must bisect $\angle ACB$. Hence, $\angle OCP = \angle ACB/2 = 30^\circ$ and $\triangle OCP$ is a 30-60-90 triangle. Since P is the midpoint of $\overline{BC}$, we have $CP = 3$, so

$$OC = CP \left(\frac{2}{\sqrt{3}} \right) = \frac{6}{\sqrt{3}} = 2\sqrt{3}.$$

□

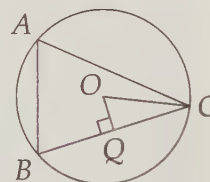
**Exercises**

7.2.1 Where is the circumcenter of an acute triangle – in, on, or outside the triangle? How about an obtuse triangle? (You do not need to prove your answer – you'll learn tools later that will make these proofs easier.)

7.2.2 Find the radius of the shown circle with center O given that $BC = 8$ and $QO = 2$.

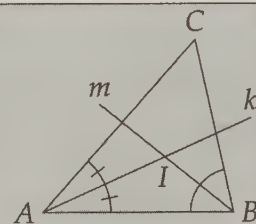
7.2.3 Find the circumradius of an equilateral triangle with side length 18.

7.2.4 What is the circumradius of a right triangle with legs of length 6 and 8?

**7.3 Angle Bisectors of a Triangle****Problems**

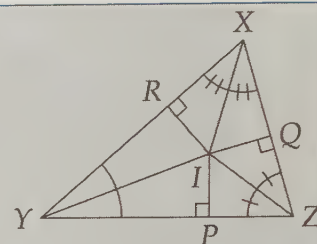
Problem 7.9: Let k be the angle bisector of $\angle CAB$ and m be the angle bisector of $\angle ABC$. Let k and m meet at I .

- Every point on k must be equidistant from what two lines?
- Every point on m must be equidistant from what two lines?
- Why must I be on the angle bisector of $\angle C$?



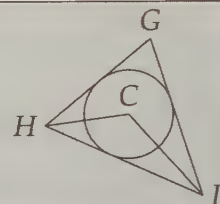
Problem 7.10: The angle bisectors of $\triangle XYZ$ meet at I , and perpendicular segments from I to the sides of the triangle are drawn as shown.

- Why is $IR = IQ = IP$?
- Suppose we draw a circle with center I and radius IP . Does the circle hit $\overline{YZ}$ at any point besides P ?
- Show that the circle from (b) is tangent to sides $\overline{XZ}$ and $\overline{XY}$ at Q and R , respectively.



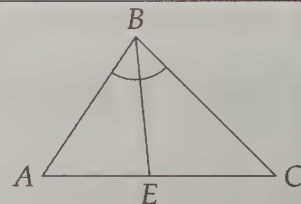
Problem 7.11: Segments $\overline{GH}$, $\overline{HI}$, and $\overline{IG}$ are tangent to the circle with center C as shown. Given that $\angle GHI = 70^\circ$ and $\angle GIH = 50^\circ$, find the following:

- $\angle CIH$.
- $\angle HCI$.

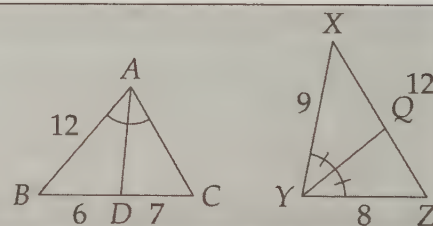


Problem 7.12: In the diagram, $\overline{BE}$ is the angle bisector of $\angle B$. In this problem, we will prove the Angle Bisector Theorem, which states that if $\overline{BE}$ is an angle bisector as shown, then $AB/AE = CB/CE$.

- Ratios make us think of similar triangles, which make us think of parallel lines. Draw a line through C parallel to $\overline{AB}$, then extend $\overline{BE}$ past E to hit this new line at X . What triangles are similar?
- What type of triangle is $\triangle BCX$?
- Use your similar triangles from (a) and your observation in (b) to prove that $AB/AE = CB/CE$.

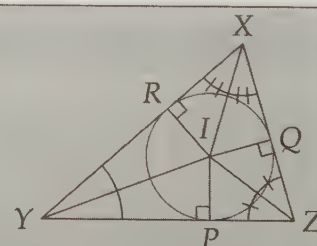


Problem 7.13: Find AC and XQ in the triangles shown. ($XZ = 12$)



Problem 7.14: Back in Problem 7.10, we discovered a circle that is tangent to all three sides of $\triangle XYZ$. In this problem, we will discover a formula for the area of $\triangle XYZ$ in terms of the radius of this circle and the perimeter of the triangle.

- Break $\triangle XYZ$ into three triangles: $\triangle XYI$, $\triangle YZI$, and $\triangle ZXI$. Find the area of each triangle in terms of one of the sides of $\triangle XYZ$ and r , the radius of the circle in the diagram.
- Find the desired formula for $[XYZ]$ by adding $[XYI]$, $[YZI]$, and $[ZXI]$.



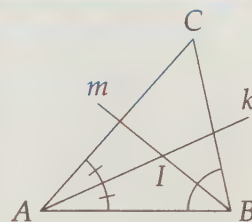
Problem 7.15: In this problem we find the radius of a circle that is tangent to all three sides of $\triangle ABC$ given that the sides of $\triangle ABC$ have lengths 7, 24, and 25.

- What kind of triangle is $\triangle ABC$? What are the area and perimeter of $\triangle ABC$?
- What is the radius of the circle tangent to all three sides of $\triangle ABC$?

Having found that the perpendicular bisectors of the sides of a triangle are concurrent, we wonder if a similar proof can be used to show that the bisectors of the angles of a triangle are also concurrent.

Problem 7.9: Prove that the angle bisectors of a triangle are concurrent.

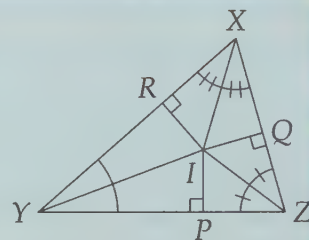
Solution for Problem 7.9: We start much as we did with the perpendicular bisectors. We consider the intersection of two of our angle bisectors, point I in the diagram. Since I is on the bisector of $\angle CAB$, it is equidistant from $\overrightarrow{AC}$ and $\overrightarrow{AB}$. Since it is on the angle bisector of $\angle ABC$, it is equidistant from $\overrightarrow{AB}$ and $\overrightarrow{BC}$. Therefore, I is equidistant from all three sides of the triangle. Specifically, since I is equidistant from $\overrightarrow{AC}$ and $\overrightarrow{BC}$, I is also on the angle bisector of $\angle ACB$, as we proved in Problem 7.4. $\square$



We found a special circle centered at the intersection of the perpendicular bisectors. Let's see if the intersection of the angle bisectors is the center of a special circle, too!

Problem 7.10: The angle bisectors of $\triangle XYZ$ meet at I , and perpendicular segments from I to the sides of the triangle are drawn as shown.

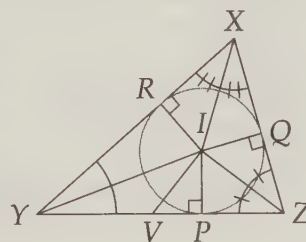
- Why is $IR = IQ = IP$?
- Suppose we draw a circle with center I and radius IP . Does the circle hit $\overline{YZ}$ at any point besides P ?
- Show that the circle from (b) is tangent to sides $\overline{XZ}$ and $\overline{XY}$ at Q and R , respectively.



Solution for Problem 7.10:

- In the previous problem, we found that I , the intersection of the angle bisectors of $\triangle XYZ$, is equidistant from the sides of $\triangle XYZ$. Since IR , IQ , and IP are the distances from I to the sides of $\triangle XYZ$, we have $IR = IQ = IP$.
- One way we can show that the circle with center I and radius IP doesn't hit $\overline{YZ}$ anywhere else is to show that all the other points on $\overline{YZ}$ are *outside* the circle.

As shown, point V is a point on $\overline{YZ}$ besides point P . From right triangle $\triangle IPV$, we have $IV^2 = IP^2 + VP^2$, so $IV > IP$. Our circle consists of all points exactly IP from I . Since V is farther than IP from I , it must be outside our circle. There's nothing special about V ; all points on $\overline{YZ}$ besides P are outside our circle. Therefore, it is impossible for the circle with radius IP to hit $\overline{YZ}$ at a second point. So, the circle must be tangent to $\overline{YZ}$.



- (c) Since $IP = IQ = IR$, the circle with center I and radius IP goes through Q and R , too. For the same reason as in the previous part, this circle is tangent to all three sides of $\triangle XYZ$.

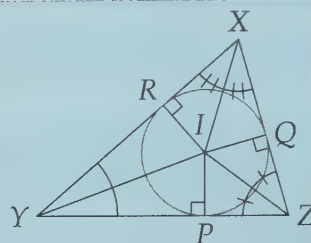
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We have found the center of a special circle at the intersection of the angle bisectors of a triangle.

Important:

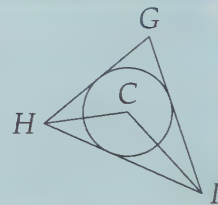


The angle bisectors of a triangle are concurrent at a point called the **incenter**. This point is equidistant from the sides of the triangle. This common distance from the incenter to the sides of a triangle is called the **inradius**, because the circle with center I and this radius is tangent to all three sides of the triangle. This circle is unsurprisingly called the **incircle** because it is inscribed in the triangle (meaning it is tangent to all the sides of the triangle). The incenter is usually denoted I , and the inradius is usually written as r .



Each triangle has exactly one incircle. (You'll be asked to prove this in Problem 12.43.) Therefore, if a circle is tangent to all three sides of a triangle, its center is the intersection of the angle bisectors of the triangle.

Problem 7.11: Segments $\overline{GH}$, $\overline{HI}$, and $\overline{IG}$ are tangent to the circle with center C as shown. Given that $\angle GHI = 70^\circ$ and $\angle GIH = 50^\circ$, find $\angle HCI$.



Solution for Problem 7.11: Since the circle is tangent to all three sides of $\triangle GHI$, it is the incircle of $\triangle GHI$. Therefore, C is the incenter of $\triangle GHI$, so $\overline{CI}$ and $\overline{CH}$ are bisectors of angles $\angle GIH$ and $\angle GHI$, respectively.

$\angle CHI = (\angle GHI)/2 = 35^\circ$ and $\angle CIH = (\angle GIH)/2 = 25^\circ$, so we have $\angle HCI = 180^\circ - 35^\circ - 25^\circ = 120^\circ$. □

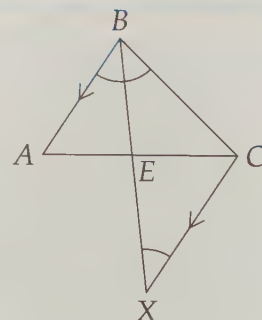
The incenter and the incircle aren't the only useful aspects of angle bisectors.

Problem 7.12: Given that E is on $\overline{AC}$ such that $\overline{BE}$ is the bisector of $\angle ABC$, prove that $AB/AE = CB/CE$.

Solution for Problem 7.12: Seeing ratios, we think of similar triangles. Similar triangles make us think of parallel lines. We draw a line through C parallel to $\overline{AB}$ and we extend $\overline{BE}$ past E to hit this new line at X . We pick this parallel line to draw because it lets us use the angles that are formed by the angle bisector. Specifically, we have $\angle BXC = \angle ABX$. $\angle ABX$ also equals $\angle XBC$, so $\triangle XCB$ is isosceles and $CB = CX$.

Our parallel line gives us similar triangles: $\triangle ABE \sim \triangle CXE$. This similarity gives us $AB/AE = CX/CE$. Since $CB = CX$, we have the desired $AB/AE = CB/CE$.

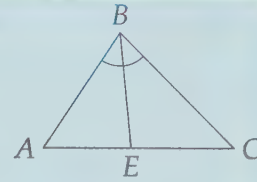
□



Important: The **Angle Bisector Theorem** states that if E is on $\overline{AC}$ such that $\overline{BE}$ is the angle bisector of $\angle B$ in triangle $\triangle ABC$, then

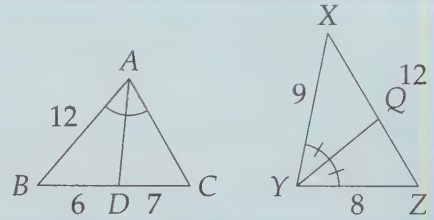
$$\frac{AB}{AE} = \frac{CB}{CE}.$$

We should think of this any time we have a problem involving lengths and angle bisectors.



For example, try using the Angle Bisector Theorem on this problem.

Problem 7.13: Find AC and XQ in the diagram. ($XZ = 12$)



Solution for Problem 7.13: From the Angle Bisector Theorem applied to $\triangle ABC$, we have $AB/BD = AC/CD$, so $12/6 = AC/7$ and $AC = 14$.

Finding XQ is a little more challenging. We let $XQ = x$, so $QZ = 12 - x$. The Angle Bisector Theorem tells us

$$\frac{XY}{XQ} = \frac{ZY}{ZQ},$$

so we have

$$\frac{9}{x} = \frac{8}{12 - x}.$$

Cross-multiplying gives $9(12 - x) = 8x$, so $x = 108/17$. Therefore, $XQ = x = 108/17$. Note that $XQ = (YX)(XZ)/(YX + YZ)$. Is this a coincidence? $\square$

But wait, there's more. We can even relate an incircle to the area of its triangle!

Problem 7.14: Find the area of $\triangle XYZ$ in terms of its inradius and its side lengths.

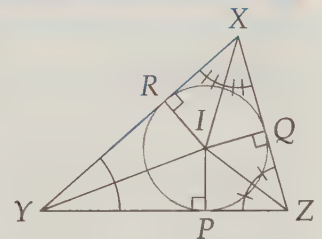
Solution for Problem 7.14: Those radii of the incircle perpendicular to the sides make us think of altitudes. Indeed, if we break $\triangle XYZ$ into three triangles with I as a vertex, these radii are altitudes. Adding the areas of these three triangles gives us $[XYZ]$:

$$[XYZ] = [XYI] + [YZI] + [XZI] = \frac{(IR)(XY)}{2} + \frac{(IP)(YZ)}{2} + \frac{(IQ)(XZ)}{2}.$$

Since $IR = IP = IQ = r$, the inradius, and $XY + YZ + XZ = 2s$, where s is the semiperimeter of $\triangle XYZ$, we have:

$$[XYZ] = \frac{r(XY)}{2} + \frac{r(YZ)}{2} + \frac{r(XZ)}{2} = r \left(\frac{XY + YZ + XZ}{2} \right) = rs.$$

$\square$



Important: The area of a triangle equals its inradius times its semiperimeter.



This $[ABC] = rs$ can be a useful tool in problems involving the area of a triangle or the inradius (or both).

Problem 7.15: Find the radius of a circle that is tangent to all three sides of $\triangle ABC$ given that the sides of $\triangle ABC$ have lengths 7, 24, and 25.

Solution for Problem 7.15: Since $7^2 + 24^2 = 25^2$, the triangle is a right triangle with legs of length 7 and 24. (This problem is an example of why it's useful to recognize Pythagorean triples.) So, we can easily find $[ABC] = (7)(24)/2 = 84$. Since we can also find the semiperimeter, we can find the inradius. Because $s = (7 + 24 + 25)/2 = 28$, we have

$$r = \frac{[ABC]}{s} = \frac{84}{28} = 3.$$

□

Exercises

7.3.1 $\overline{AB}$, $\overline{AC}$, and $\overline{BC}$ are tangent to $\odot O$. If $\angle A = 70^\circ$, $\angle B = 72^\circ$, and $\angle C = 38^\circ$, what is the measure of $\angle OAB$? **Hints:** 228

7.3.2 Is the incenter of an obtuse triangle inside, outside, or on the triangle?

7.3.3 Can the incenter and the circumcenter of a triangle ever be the same point?

7.3.4 $\overline{PS}$ and $\overline{RT}$ are angle bisectors of $\triangle PQR$, and $\angle PIR = 130^\circ$.

(a) Given $\angle RPQ = 30^\circ$, find $\angle PRT$, $\angle PRQ$, and $\angle Q$.

(b) Given $\angle RPQ = 50^\circ$, find $\angle PRT$, $\angle PRQ$, and $\angle Q$.

(c) Given $\angle RPQ = 80^\circ$, find $\angle PRT$, $\angle PRQ$, and $\angle Q$.

(d) Do you notice anything unusual in your answers for $\angle Q$? Why does this unusual pattern occur?

7.3.5 Find RZ in the figure at right if $PR = 9$, $QZ = 4$, $PQ = 6$, and $\overline{PZ}$ bisects $\angle QPR$.

7.3.6 In $\triangle ABC$, $AB = 10$, $AC = 12$, and $BC = 8$. Point M is on side $\overline{BC}$ such that $\angle BAM = \angle CAM$. Find BM .

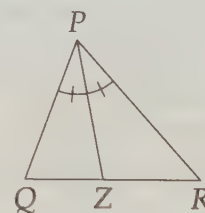
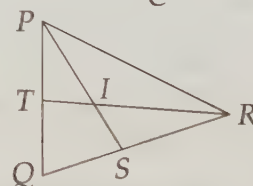
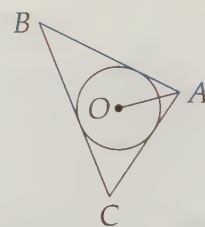
7.3.7 For each part below, find the length of the inradius of a triangle with the given numbers as side lengths.

(a) 3, 4, 5

(b) 6, 6, 6

(c) 7, 7, 10

(d) 5, 6, 7



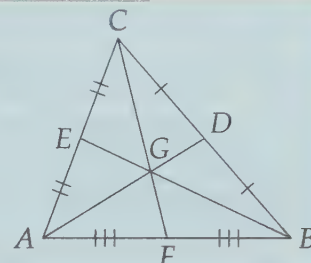
7.4 Medians

A **median** of a triangle is a segment from a vertex to the midpoint of the opposite side. In the figure below, $\overline{AD}$, $\overline{BE}$, and $\overline{CF}$ are all medians.

Important:



The medians of a triangle are concurrent at a point called the **centroid** of the triangle. The centroid of the triangle is usually labeled G .

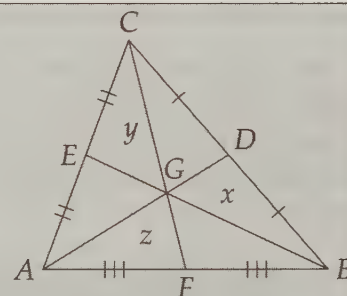


In each of the problems in this section, you can assume the medians of a triangle are concurrent. We'll prove that the medians of a triangle are concurrent in Chapter 17. You'll also have a chance to do it yourself in the Challenge Problems. (Hint: We suggest re-reading Section 4.3.)

Problems

Problem 7.16: In this problem we will prove that the medians of a triangle divide the triangle into 6 triangles of equal area. We will make heavy use of the same base/same altitude principle we studied in Section 4.3.

- Let $[DGB] = x$, $[CGE] = y$, and $[AGF] = z$ as shown in the diagram. Prove that $[DGC] = x$, $[AGE] = y$, and $[BGF] = z$.
- Show that $[ABD] = [ACD]$ and use this to prove that $y = z$.
- Prove that the six triangles formed by drawing the medians of $\triangle ABC$ have equal area.



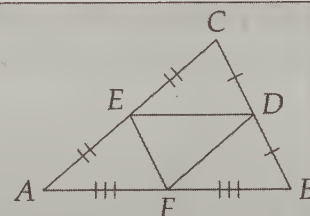
Problem 7.17: Use triangles $\triangle AGB$ and $\triangle DGB$ in the previous problem to prove that the centroid cuts each median in a 2 : 1 ratio, specifically $AG/GD = 2$.

Problem 7.18: $\triangle PQR$ is an equilateral triangle with side length 12.

- Show that each angle bisector of an equilateral triangle is also a median.
- Find the length of the inradius of $\triangle PQR$.

Problem 7.19: Points D , E , and F are the midpoints of the sides of $\triangle ABC$ as shown below.

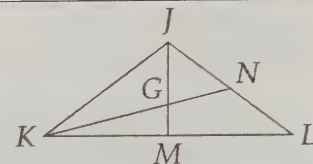
- Show that $\triangle BDF \sim \triangle BCA$.
- Show that $\triangle DEF \sim \triangle ABC$.
- Show that $\overline{EF} \parallel \overline{CB}$.
- Show that each of the four small triangles in the diagram has area equal to $1/4$ the area of $\triangle ABC$.



Problem 7.20: $\triangle ABC$ has sides of length $AB = 9$, $BC = 12$, and $CA = 15$. Find the length of the median from B to the midpoint of $\overline{AC}$. **Hints:** 229, 537

Problem 7.21: In $\triangle JKL$, $JK = JL = 10$ and $KL = 16$.

- Find the length of median $\overline{JM}$.
- Let medians $\overline{JM}$ and $\overline{KN}$ meet at G . Find KG .
- Find the length of median $\overline{KN}$.

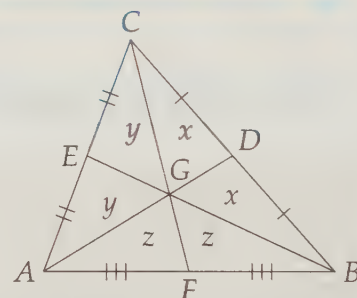


Medians create triangles with equal bases; therefore, they create triangles with equal areas. We can use these equal areas to learn even more about the medians of a triangle.

Problem 7.16: Prove that the medians of a triangle divide the triangle into six triangles with equal area.

Solution for Problem 7.16: Let our triangle be $\triangle ABC$ with midpoints D , E , and F , and with centroid G , as shown. We have $[BGD] = [CGD]$ because these triangles have the same altitude from G to equal bases $\overline{BD}$ and $\overline{CD}$. Similarly, $[AGE] = [CGE]$ and $[AFG] = [BFG]$. We assign variables to these areas as shown in the diagram at right.

The little triangles aren't the only triangles that share altitudes. Since $\triangle ABD$ and $\triangle ACD$ have the same altitude from A to equal bases $\overline{BD}$ and $\overline{CD}$, we have $[ABD] = [ACD]$. Therefore, $x + 2z = x + 2y$, so $y = z$. Similarly, we can use $[AFC] = [BFC]$ to show that $x = y$, so we have $x = y = z$ and all six little triangles have the same area. $\square$



Important: The medians of a triangle divide the triangle into six little triangles of equal area.



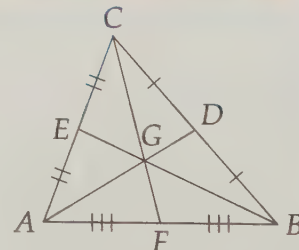
We can use this property of medians to discover an interesting property of the centroid.

Problem 7.17: Prove that the centroid of any triangle cuts each of the triangle's median in a 2 : 1 ratio, with the longer portion being the segment from the centroid to the vertex.

Solution for Problem 7.17: Our work with areas gives us the solution right away. Since $\triangle AGB$ consists of two of the little equal-area triangles and $\triangle GDB$ is only one of them, we have $[AGB] = 2[GDB]$. These two triangles share an altitude from B to bases $\overline{AG}$ and $\overline{GD}$, so

$$\frac{AG}{GD} = \frac{[AGB]}{[GDB]} = 2.$$

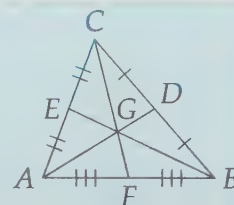
Clearly this works for any median. $\square$



Important:

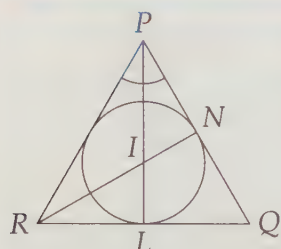
The centroid of a triangle cuts its medians into a 2 : 1 ratio. For example, for the triangle shown, we have

$$\frac{AG}{GD} = \frac{BG}{GE} = \frac{CG}{GF} = \frac{2}{1}.$$



This simple fact can be useful in a variety of problems. Here's an example.

Problem 7.18: $\triangle PQR$ is an equilateral triangle with side length 12. Find the length of the inradius of $\triangle PQR$.



Solution for Problem 7.18: Let $\overline{PL}$ and $\overline{RN}$ be angle bisectors of $\triangle PQR$. Since the angle bisectors of an equilateral triangle are also the medians of the triangle, L is the midpoint of $\overline{RQ}$ and the incenter, I , is also the centroid. So, I divides median $\overline{PL}$ such that $PI/IL = 2/1$. Therefore, $IL = PL/3$.

$\triangle PLR$ is a 30-60-90 triangle, so $RL = PR/2 = 6$ and $PL = RL\sqrt{3} = 6\sqrt{3}$. Finally, we have $IL = PL/3 = 2\sqrt{3}$. (Note: We could also have solved this problem by noting that $\triangle IRL$ is a 30-60-90 triangle.) $\square$

Instead of connecting the midpoints to the opposite vertices, suppose we connect them to each other, as shown in Figure 7.3. $\triangle DEF$ is called the **medial triangle** of $\triangle ABC$.

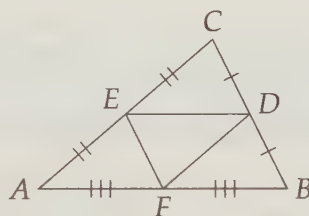


Figure 7.3: Medial Triangle

Problem 7.19: Prove that the four small triangles in Figure 7.3 are congruent, and that each is similar to $\triangle ABC$.

Solution for Problem 7.19: $\triangle FAE$ and $\triangle BAC$ have an angle in common, but we don't know anything about the other angles of the triangles. So, we look to SAS Similarity to prove similarity. Since $AF/AB = AE/AC = 1/2$, we have $\triangle FAE \sim \triangle BAC$ by SAS Similarity. Similarly, we can show $\triangle FBD \sim \triangle ABC$ and $\triangle CED \sim \triangle CAB$.

Since each of the small triangles above have lengths that are $1/2$ the sides of $\triangle ABC$, we have

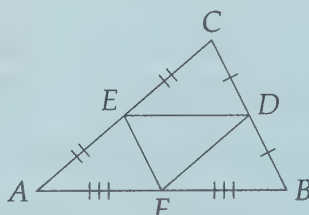
$$\frac{FE}{BC} = \frac{DE}{AB} = \frac{DF}{AC} = \frac{1}{2}.$$

Therefore, $\triangle DEF \sim \triangle ABC$ by SSS Similarity.

Since all four little triangles are similar to $\triangle ABC$ with the same side ratio, the four little triangles must be congruent. $\square$

Notice that we can also use $\triangle AFE \sim \triangle ABC$ to show that $\angle AFE = \angle ABC$, from which we have $\overline{FE} \parallel \overline{BC}$. Similarly we can show that the other two sides of the medial triangle are parallel to their respective sides of the original triangle.

Important: Given that $\triangle DEF$ is the medial triangle of $\triangle ABC$ as shown below, we have:



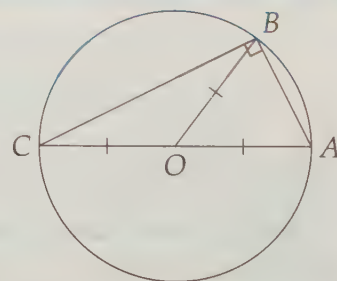
- I. $\triangle DEF \sim \triangle ABC$, $\triangle DEF \cong \triangle FBD \cong \triangle AFE \cong \triangle EDC$.
- II. $\frac{EF}{BC} = \frac{DE}{AB} = \frac{DF}{AC} = \frac{1}{2}$.
- III. $\overline{DF} \parallel \overline{AC}$, $\overline{EF} \parallel \overline{BC}$, and $\overline{DE} \parallel \overline{AB}$.

Together, facts II and III are sometimes called the **Midline Theorem**.

Let's try a couple problems using the properties of medians we have just found.

Problem 7.20: $\triangle ABC$ has sides of length $AB = 9$, $BC = 12$, and $CA = 15$. Find the length of the median from B to the midpoint of $\overline{AC}$.

Solution for Problem 7.20: First we notice that $\triangle ABC$ has side lengths in the ratio $3 : 4 : 5$, so it is a right triangle with hypotenuse $\overline{AC}$. Therefore, we are looking for the length of the median to the hypotenuse of a right triangle. Since the midpoint of the hypotenuse of a right triangle is also the circumcenter of the right triangle (see Problem 7.6), the median to the midpoint of the hypotenuse is a radius of the circumcircle as shown in the diagram. Therefore, the length of our median equals the circumradius, which is half the hypotenuse. So, our answer is $CA/2 = 15/2$. $\square$



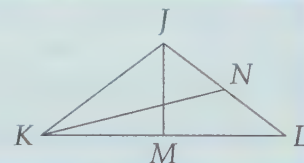
Concept: When given the lengths of the sides of a triangle in a problem, always take the time to check if it is a right triangle. Special properties of right triangles often simplify problems.

Important: The median to the hypotenuse of a right triangle is equal in length to half the hypotenuse.



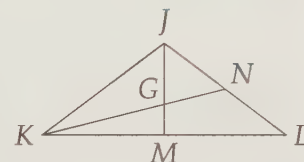
Problem 7.21: In $\triangle JKL$, $JK = JL = 10$ and $KL = 16$.

- Find the length of median $\overline{JM}$.
- Find the length of median $\overline{KN}$.



Solution for Problem 7.21:

- Since $JK = JL$, median $\overline{JM}$ is also an altitude (since $\triangle JKM \cong \triangle JLM$). Therefore, $\triangle JMK$ is a right triangle with hypotenuse $JK = 10$ and one leg of length $KM = 8$. So, $JM = 6$.
- Let $\overline{JM}$ and $\overline{KN}$ meet at G , which is the centroid of $\triangle JKL$. It's not obvious how to get KN , but we do have right triangle $\triangle KGM$, which we can use to find KG . Since G is the centroid and $\overline{JM}$ is a median, $GM = JM/3 = 2$. Therefore, $KG = \sqrt{GM^2 + KM^2} = \sqrt{4 + 64} = 2\sqrt{17}$. Since $\overline{KN}$ is a median and G the centroid, $KG/KN = 2/3$. Therefore, $KN = (3/2)KG = 3\sqrt{17}$.



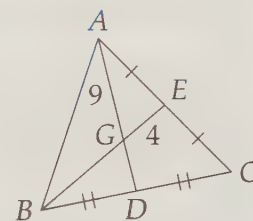
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Exercises

7.4.1 D and E are the midpoints of sides $\overline{BC}$ and $\overline{AC}$ of $\triangle ABC$ in the diagram at right. $\overline{AD}$ and $\overline{BE}$ meet at G , $AG = 9$, and $GE = 4$. Find GD and BG .

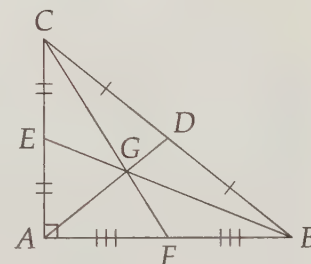
7.4.2 Right triangle $\triangle PQR$ has $PQ = 5$, $QR = 12$, and $RP = 13$. The midpoints of $\overline{QR}$, $\overline{PR}$, and $\overline{PQ}$ are A , B , and C , respectively.

- Draw a diagram. Mark the right angle.
- Find the length of median $\overline{QB}$.
- Find the lengths of medians $\overline{PA}$ and $\overline{RC}$.



7.4.3 The three medians of right triangle $\triangle ABC$ intersect at G as shown. Given that $AC = 8$ and $AB = 15$, find the following:

- $[ABC]$.
- $[AFC]$.
- $[ACD]$.
- $[AEG]$.
- $[EGDC]$.
- $[AFGE]$.
- $[DEF]$.
- $[BFD]$.
- ★ CG . **Hints:** 403



7.4.4★ Medians $\overline{XA}$ and $\overline{YB}$ of $\triangle XYZ$ have the same length. Prove that $XZ = YZ$. **Hints:** 167, 375

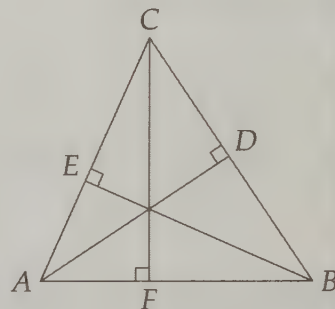
7.5 Altitudes

We've already seen altitudes in our investigation of triangle area. You can probably guess by now what we'll prove first about altitudes.

Problems

Problem 7.22: In this problem we will prove that the lines containing the altitudes of any triangle are concurrent. This is a pretty slick proof. Before trying it, you might want to review what we learned about the medial triangle on page 186.

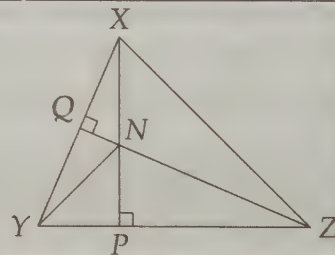
- Our diagram shows altitudes $\overline{AD}$, $\overline{BE}$, and $\overline{CF}$ of $\triangle ABC$. Draw a line through A parallel to $\overline{BC}$, a line through B parallel to $\overline{AC}$, and a line through C parallel to $\overline{AB}$. The intersections of these three lines form another triangle. Call this triangle $\triangle JKL$, with A on $\overline{KL}$ and B on $\overline{JL}$.
- Show that $\triangle CAK \cong \triangle ACB$.
- Use part (b) and similar congruences to prove that A , B , and C are the midpoints of $\overline{KL}$, $\overline{JL}$, and $\overline{JK}$, respectively.
- How are lines $\overleftrightarrow{AD}$, $\overleftrightarrow{BE}$, and $\overleftrightarrow{CF}$ related to $\triangle JKL$?
- Prove that the altitudes of $\triangle ABC$ are concurrent.



Problem 7.23: Where do altitudes of a right triangle intersect – inside, outside, or on the triangle? How about an acute triangle? An obtuse triangle? (If necessary, extend the altitudes to the point where they all meet.)

Problem 7.24: Altitudes $\overline{QZ}$ and $\overline{XP}$ intersect at N as shown. Given that $\angle YXZ = 70^\circ$ and $\angle XZY = 45^\circ$, find each of the following:

- $\angle ZXP$.
- $\angle XZQ$.
- $\angle YXP$.
- ★ $\angle NYZ$. **Hints:** 194



Problem 7.25: The altitudes of $\triangle ABC$ meet at point H . At what point do the altitudes of $\triangle ABH$ meet? How about the altitudes of $\triangle ACH$? $\triangle BCH$? (As usual, extend the altitudes if necessary.)

We start, as you probably guessed, by proving that the altitudes of a triangle are concurrent. This is a bit of a magical proof, and it uses what we have already learned about perpendicular bisectors and the medial triangle.

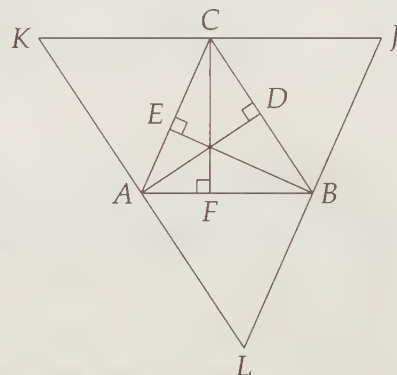
Problem 7.22: Prove that the lines containing the altitudes of a triangle are concurrent.

Solution for Problem 7.22: Our altitudes make us think of perpendicular lines. Concurrent perpendicular

lines make us think of perpendicular bisectors. Unfortunately, there's no simple solution like we had with perpendicular bisectors. So, we put our wishful thinking hat on, and wonder if we could possibly show that the altitudes of our triangle are also the perpendicular bisectors of the sides of some other triangle. If so, we would know they are concurrent, since the perpendicular bisectors of our 'other triangle' must be concurrent. But what's our 'other triangle'?

We want our A , B , and C to be the midpoints of the sides of our other triangle (since then the altitudes of $\triangle ABC$ would be perpendicular bisectors of the sides of the other triangle). This means we want $\triangle ABC$ to be the medial triangle of some other triangle. Since we know that the sides of a triangle are parallel to the sides of its medial triangle, we know how to make a triangle starting with its medial triangle.

We draw lines $\overleftrightarrow{JK}$, $\overleftrightarrow{KL}$, and $\overleftrightarrow{LJ}$ through the vertices of $\triangle ABC$ parallel to the sides of $\triangle ABC$ as shown. To prove that $\triangle ABC$ is indeed the medial triangle of $\triangle JKL$, we must show that the vertices of $\triangle ABC$ are the midpoints of the sides of $\triangle JKL$. Since $\overline{AB} \parallel \overline{JK}$, $\angle CAB = \angle ACK$. Similarly, $\angle CAK = \angle BCA$, so $\triangle CAK \cong \triangle ACB$. Likewise, $\triangle BLA \cong \triangle ACB$, so $\triangle BLA \cong \triangle CAK$. Therefore, $AL = AK$, so A is the midpoint of $\overline{LK}$.



In the same way, we find that B and C are the midpoints of $\overline{LJ}$ and $\overline{JK}$, respectively. Therefore, $\triangle ABC$ is the medial triangle of $\triangle JKL$. Since $\overline{KL} \parallel \overline{BC}$, altitude $\overline{AD}$ of $\triangle ABC$ is perpendicular to $\overline{KL}$. Since $\overline{AD}$ is perpendicular to $\overline{KL}$ and passes through its midpoint, it is the perpendicular bisector of $\overline{KL}$. Similarly, $\overline{CF}$ and $\overline{BE}$ are also perpendicular bisectors of the sides of $\triangle JKL$. Lines $\overline{AD}$, $\overline{BE}$, and $\overline{CF}$ are concurrent because they are the perpendicular bisectors of the sides of $\triangle JKL$. These lines also contain the altitudes of $\triangle ABC$, so we have proved that the lines containing the altitudes of $\triangle ABC$ are concurrent. $\square$

Important: The altitudes of any triangle are concurrent at a point called the **orthocenter**. We usually denote the orthocenter with the letter H .



To get a feel for the orthocenter, we look for it in a few different types of triangles.

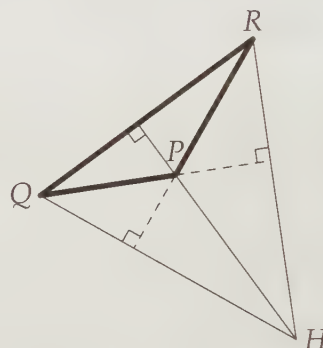
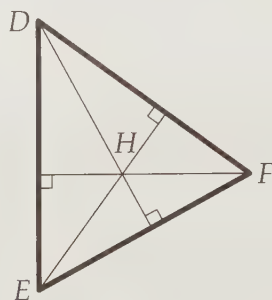
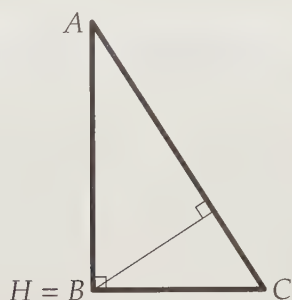
Problem 7.23: Where do altitudes of a right triangle intersect – inside, outside, or on the triangle? How about an acute triangle? An obtuse triangle?

Extra! Euclid's *Elements* used to be one of the cornerstones of a classical education, and was used by many self-educated people to hone their logic skills. Noted Abraham Lincoln biographer Carl Sandburg wrote that Lincoln 'bought the *Elements* of Euclid...[It] went into his carpetbag as he went out on the circuit. At night...he read Euclid by the light of a candle after others had dropped off to sleep.' (Source: *Journey Through Genius* by William Dunham)

You can follow in Lincoln's footsteps by reading Euclid's *Elements* online. A link to the online book is provided on our Links page described in the Website section on page viii.

CHAPTER 7. SPECIAL PARTS OF A TRIANGLE

Solution for Problem 7.23: We start with right triangle $\triangle ABC$ with right angle at B . Point B is the foot of the altitude from A to $\overline{BC}$ and the foot of the altitude from C to $\overline{AB}$; therefore, the orthocenter of a right triangle is the vertex of the right angle.



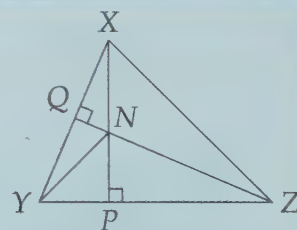
In acute $\triangle DEF$, we see that all the altitudes go inside the triangle, so the orthocenter is inside the triangle.

In obtuse $\triangle PQR$, the altitudes from the two acute angles are entirely outside the triangle (except where they pass through a vertex). Therefore, the orthocenter of an obtuse triangle is outside the triangle. $\square$

Let's try using the orthocenter in a problem.

Problem 7.24: Altitudes $\overline{QZ}$ and $\overline{XP}$ of $\triangle XYZ$ intersect at N . Given that $\angle YXZ = 70^\circ$ and $\angle XZY = 45^\circ$, find the following:

- (a) $\angle ZXP$.
- (b) $\angle XZQ$.
- (c) $\angle YXP$.
- (d)★ $\angle NYZ$.



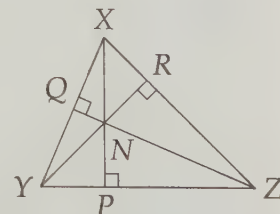
Solution for Problem 7.24:

- (a) From right triangle $\triangle XPZ$, we have $\angle ZXP = 90^\circ - \angle XZP = 45^\circ$.
- (b) From right triangle $\triangle XZQ$, we have $\angle XZQ = 90^\circ - \angle ZXQ = 90^\circ - \angle ZXY = 20^\circ$.
- (c) We could use either right triangle $\triangle XYP$, or we could use $\angle ZXP$ and $\angle YXZ$:

$$\angle YXP = \angle YXZ - \angle ZXP = 70^\circ - 45^\circ = 25^\circ.$$

- (d) Since two altitudes meet at N , we know that N is the orthocenter. Therefore, if we continue $\overline{YN}$ to meet $\overline{XZ}$ at R , we know that $\overline{YR}$ is also an altitude. From right triangle $\triangle RYZ$, we find that $\angle NYZ = 90^\circ - \angle RZY = 45^\circ$.

$\square$



Notice that recognizing the orthocenter was a key step in the solution to our last part. This is just one small example of how the awareness of the special points, lines, and circles of a triangle can help.

Concept:

If we have the altitudes from two vertices of a triangle, then we know that the line through the intersection point of these two altitudes and the third vertex is also an altitude. Once you have two altitudes, you should almost always think about this third altitude.

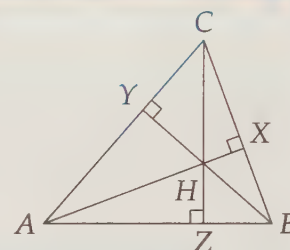
This, of course, also works with our other special lines – if you have two medians, you have the third; if you have two angle bisectors, you have the third, etc. With these other special lines you also get a little more, since we have special knowledge about the centroid (the 2 : 1 ratio), the incenter (center of the inscribed circle), and the circumcenter (center of the circumscribed circle).

Again, if you have two of any of these special lines, you should consider the third line and remember what you know about the intersection point.

Here's an interesting tidbit about altitudes and orthocenters:

Problem 7.25: The altitudes of $\triangle ABC$ meet at point H . At what point do the altitudes of $\triangle ABH$ meet? How about $\triangle ACH$? $\triangle BCH$?

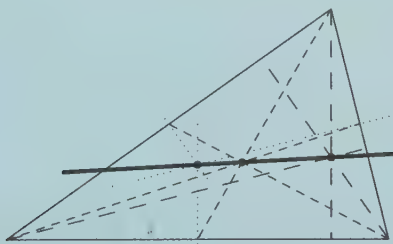
Solution for Problem 7.25: Our diagram shows that the altitudes of $\triangle ABH$ are $\overleftrightarrow{HZ}$ (perpendicular to side $\overline{AB}$), $\overleftrightarrow{AC}$ (perpendicular to $\overleftrightarrow{BH}$), and $\overleftrightarrow{BC}$ (perpendicular to $\overleftrightarrow{AH}$). These lines clearly all pass through C , so C is the orthocenter of $\triangle ABH$. Similarly, B is the orthocenter of $\triangle AHC$ and A is the orthocenter of $\triangle BHC$. $\square$



And here's an interesting fact about how the orthocenter, circumcenter, and centroid of a triangle are related:

Sidenote:

In any triangle, the centroid, G , is on the line segment connecting the orthocenter, H , to the circumcenter, O , such that $2OG = GH$. This line through these three points is called the **Euler line**, after the great mathematician **Leonhard Euler**. One such line is shown below.

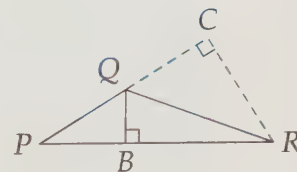


Euler was one of the most prolific mathematicians ever, producing notable work in nearly every field of mathematics, and creating new fields himself. The Euler line is just one of many mathematical results that have been named after Euler. His impact on mathematics was so great that much of our notation today was inspired by him, such as $f(x)$ for functions, i for $\sqrt{-1}$, e for the base of common logs, Σ for summation, and even π for, well, pi. He linked three of these with his famous statement $e^{\pi i} = -1$.

Exercises

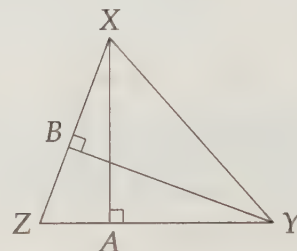
7.5.1 $\overline{RC}$ and $\overline{QB}$ are altitudes of $\triangle PQR$ as shown, $\angle QRP = 20^\circ$, and $\angle QPR = 32^\circ$. Find the following:

- (a) $\angle PQB$.
- (b) $\angle CRP$.
- (c) $\angle CQR$.
- (d) $\angle CQB$.



7.5.2 In the diagram, $YZ = 5$, $YB = 3$, and $XA = 4$.

- (a) Find $[XYZ]$.
- (b) Find BZ .
- (c) Find XZ .
- (d) Find XB .
- (e) Find XY .



7.5.3 $\triangle ABC$ is isosceles with $AB = AC$. Must the altitudes from B and C have the same length? Why or why not?

7.5.4 Why must all the altitudes of an equilateral triangle have the same length?

7.5.5 Altitude $\overline{AD}$ of $\triangle ABC$ is also an angle bisector of $\triangle ABC$. Must D be the midpoint of $\overline{BC}$? Why or why not?

7.6★ Challenging Problems

In this section we tackle several challenging problems in which we use what we've learned in this chapter about the special points, lines, and circles of a triangle. You may wish to try your hand at the Review Problems at the end of this chapter before attempting the problems in this section.

Problems

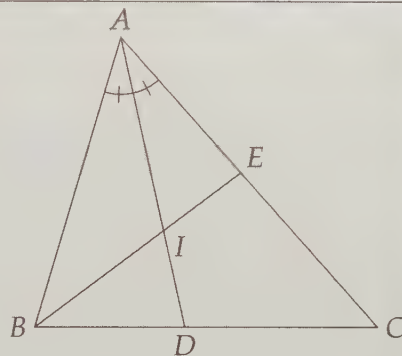
Problem 7.26: In this problem, our goal is to find the area of $ABCD$, given that $AB = BC = 6$, $CD = 3\sqrt{2}$, $\angle ABC = 90^\circ$, and that there is a circle passing through all four vertices of $ABCD$.

- (a) Where is the center, O , of the circle that passes through all four vertices?
- (b) What is the radius of this circle?
- (c) What kind of triangle is $\triangle OCD$?
- (d) What are the angles in $\triangle OAD$? What is $\angle CDA$?
- (e) Find the area of $ABCD$.

Problem 7.27: Given $AB = BC = 10$ and $AC = 12$, find the circumradius and the inradius of $\triangle ABC$.
Hints: 39, 322, 549

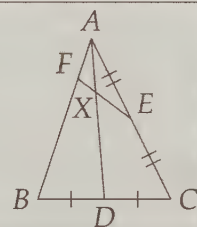
Problem 7.28: In the diagram, $\overline{AD}$ bisects $\angle BAC$ and I is the incenter of $\triangle ABC$. Furthermore, $AB = 7$, $BC = 8$, and $AC = 11$. Find AI/ID .

- Use the Angle Bisector Theorem to find another pair of segments that have a ratio equal to AI/ID .
- Use the Angle Bisector Theorem again to find another pair of lengths (besides AI and ID) that have a ratio equal to the ratio you found in part (a).
- Let $BD = x$. Set up and solve an equation for x using the ratios you found in the previous two parts. Then finish the problem by finding AI/ID .

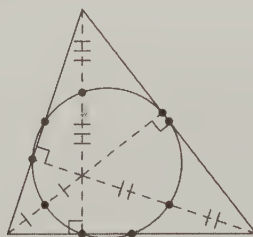


Problem 7.29: Is it possible for there to be points E and F on side $\overline{BC}$ of $\triangle ABC$ such that $BE = EF = FC$ and $\angle BAE = \angle EAF = \angle FAC$? Why or why not? **Hints:** 377

Problem 7.30: Point E is the midpoint of $\overline{AC}$, and $\overline{AD}$ is a median of $\triangle ABC$. F is on $\overline{AB}$ such that $AF = AB/4$. $\overline{EF}$ and $\overline{AD}$ meet at X . Find AX/AD . **Hints:** 95



Extra! The incircle and the circumcircle are not the only notable circles that can be found in a triangle. For any triangle, a single circle passes through the midpoints of the sides, the feet of the altitudes, and the midpoints of the segments connecting the orthocenter to the vertices. This circle is cleverly called the **nine-point circle** of the triangle.



The nine-point circle is tangent to the incircle, has a radius equal to half the circumradius (you have the tools to prove this—see if you can!), and its center is the midpoint of the segment connecting the orthocenter and the circumcenter, which we discussed on page 191. We'll prove the existence of the nine-point circle, and some of its properties, during our deeper investigations of triangles in *Intermediate Geometry*.

Problem 7.26: Find the area of $ABCD$, given that $AB = BC = 6$, $CD = 3\sqrt{2}$, $\angle ABC = 90^\circ$, and that there is a circle passing through all four vertices of $ABCD$.

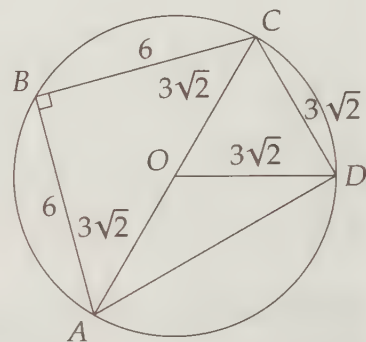
Solution for Problem 7.26: We don't know much about figures with four sides at this point, but we now know a whole lot about triangles. We start with right triangle $\triangle ABC$. The area of this triangle is $(AB)(BC)/2 = 18$, so all we have to find to finish is $[ACD]$. Unfortunately, we don't have much information about $\triangle ACD$.

We do, however, know that the circumcenter of $\triangle ABC$ is the midpoint of $\overline{AC}$. Therefore the center of the circle, O , is the midpoint of $\overline{AC}$. Breaking the problem into triangles worked well once, so we try again by drawing $\overline{OD}$. Since $\triangle ABC$ is an isosceles right triangle, $AC = 6\sqrt{2}$. So, the radius of the circle is $3\sqrt{2}$. We label all the lengths we know, and we see that $\triangle OCD$ is an equilateral triangle.

Since $\triangle OCD$ is equilateral, its angles are all 60° . Furthermore, $\angle AOD = 180^\circ - \angle COD = 120^\circ$. Since $\triangle AOD$ is isosceles, $\angle OAD = \angle ODA = 30^\circ$. Now, we see that $\triangle ACD$ is a 30-60-90 triangle, so $AD = CD\sqrt{3} = 3\sqrt{6}$. Finally, $[ACD] = (CD)(AD)/2 = 9\sqrt{3}$.

Finally, we have $[ABCD] = [ABC] + [ACD] = 18 + 9\sqrt{3}$. In Section 12.1, we will learn a much faster way to deduce that $\angle ADC$ is a right angle. $\square$

In this solution, we seemed to stumble on a surprising equilateral triangle and a useful right angle. We were able to find them because we kept track of everything we learned in our diagram.



Concept: Label your diagram with all the lengths and angles you can find. This will help you find relationships in the diagram that might be hard to find otherwise.

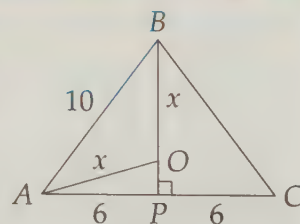
Problem 7.27: Given $AB = BC = 10$ and $AC = 12$, find the circumradius and the inradius of $\triangle ABC$.

Solution for Problem 7.27: Since we're looking for the circumradius and the inradius, we're thinking about perpendicular bisectors and angle bisectors. Our triangle is isosceles, so we'll start with the special lines from the vertex angle ($\angle B$) since they're all the same line.

$\overline{BP}$ splits our triangle in two congruent halves, so $AP = 6$. Since $\triangle BPA$ is right, we have $BP = 8$ from the Pythagorean Theorem. To find the circumradius, we build a right triangle with the circumradius as a side by connecting the circumcenter, O , to A . Since $OB = x$ and $BP = 8$, we have $OP = 8 - x$ and we can apply the Pythagorean Theorem to $\triangle OAP$ to find $OA^2 = OP^2 + AP^2$. Substituting our expressions for the sides, we have

$$x^2 = (8 - x)^2 + 36.$$

A little algebra then yields $16x = 100$, so $x = 25/4$.



Since the area of a triangle is its semiperimeter times its inradius, we can quickly find the inradius of $\triangle ABC$:

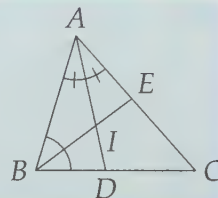
$$r = \frac{[ABC]}{s} = \frac{(AC)(BP)/2}{(10 + 10 + 12)/2} = \frac{48}{16} = 3.$$

Challenge: See if you can find the inradius using a tactic like the one we used for the circumradius! $\square$

Concept: Need to find a length in a problem? Try building right triangles.



Problem 7.28: In the diagram, $\overline{AD}$ bisects $\angle BAC$ and I is the incenter of $\triangle ABC$. Furthermore, $AB = 7$, $BC = 8$, and $AC = 11$. Find AI/ID .



Solution for Problem 7.28: We're looking for a ratio and we have an angle bisector, so we try using the Angle Bisector Theorem. Since I is the incenter of $\triangle ABC$, $\overline{BE}$ bisects $\angle ABC$. Therefore, $\overline{BI}$ is an angle bisector of $\triangle ABD$. From the Angle Bisector Theorem, we have

$$\frac{AI}{ID} = \frac{AB}{BD}.$$

We know $AB = 7$, so if we can find BD , we're finished. The Angle Bisector Theorem also gives us $AB/BD = AC/CD$ when applied to angle bisector $\overline{AD}$ of $\triangle ABC$. Since $BD + DC = BC = 8$, we have:

$$\frac{7}{BD} = \frac{11}{8 - BD}.$$

Cross-multiplying gives $7(8 - BD) = 11BD$, from which we find $BD = 28/9$. Therefore, $AI/ID = AB/BD = 9/4$. $\square$

Concept: Keep your eye on the ball! Often, working backwards from what you want will guide you to the solution.

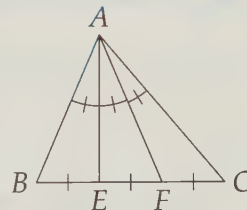


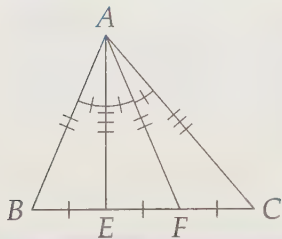
Problem 7.29: Is it possible for there to be points E and F on side $\overline{BC}$ of $\triangle ABC$ such that $BE = EF = FC$ and $\angle BAE = \angle EAF = \angle FAC$? Why or why not?

Solution for Problem 7.29: We have angle bisectors and lengths, so we try the Angle Bisector Theorem. $\overline{AE}$ and $\overline{AF}$ trisect $\angle BAC$, but we can also look at them as the angle bisectors of $\angle BAF$ and $\angle EAC$, respectively. The Angle Bisector Theorem applied to $\triangle BAF$ and $\triangle EAC$ gives us

$$\frac{AB}{AF} = \frac{BE}{EF} = 1 \quad \text{and} \quad \frac{AE}{AC} = \frac{EF}{FC} = 1.$$

Therefore, we have $AB = AF$ and $AE = EC$.





Marking these equalities in our diagram, congruent triangles jump out. Specifically, SSS tells us $\triangle ABE \cong \triangle AFE \cong \triangle AFC$. However, this means that $\angle AEB = \angle AEF = 90^\circ$ and $\angle AFE = \angle AFC = 90^\circ$. Therefore $\triangle AEF$ has two right angles, which is impossible.

So, it is not possible for there to be points E and F on side $\overline{BC}$ of $\triangle ABC$ such that $BE = EF = FC$ and $\angle BAE = \angle EAF = \angle FAC$. $\square$

Problem 7.30: Point E is the midpoint of $\overline{AC}$, and $\overline{AD}$ is a median of $\triangle ABC$. F is on $\overline{AB}$ such that $AF = AB/4$. $\overline{EF}$ and $\overline{AD}$ meet at X . Find AX/AD .

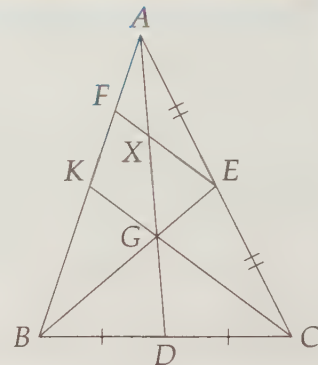
Solution for Problem 7.30: We start with a diagram. We go ahead and draw in the other two medians, hoping we can use what we know about the centroid. Since K is the midpoint of $\overline{AB}$ and $AF = AB/4$, F is the midpoint of $\overline{AK}$. So, $\triangle AFE \sim \triangle AKC$ by SAS Similarity. Therefore, $\angle AFE = \angle AKC$, so $\overline{FE} \parallel \overline{KC}$. This gives us $\triangle AFX \sim \triangle AKG$, so

$$\frac{AX}{AG} = \frac{AF}{AK} = \frac{1}{2}.$$

Since G is the centroid of $\triangle ABC$, we have $AG/AD = 2/3$, so

$$\frac{AX}{AD} = \left(\frac{AX}{AG}\right) \left(\frac{AG}{AD}\right) = \frac{1}{3}.$$

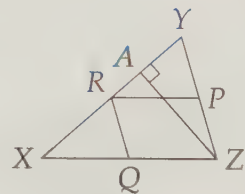
$\square$



Exercises

7.6.1 P , Q , and R are the midpoints of sides $\overline{YZ}$, $\overline{XZ}$, $\overline{XY}$, respectively, and $\overline{ZA}$ is the altitude to side $\overline{XY}$.

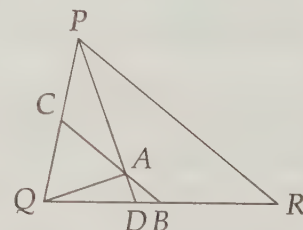
- (a) Show that $AP = RQ$. **Hints:** 298
 (b)★ Show that $\angle PRQ = \angle PAQ$. **Hints:** 47



7.6.2 In right triangle ABC , let F be the midpoint of hypotenuse $\overline{AB}$, and let D be the foot of the altitude from C to $\overline{AB}$. Let E be on $\overline{AB}$ such that $\overline{CE}$ is the angle bisector of $\angle ACB$. Prove that $\angle DCE = \angle ECF$. **Hints:** 136

7.6.3 In $\triangle PQR$, $PQ = 8$, $PR = 10$, and $QR = 9$. $\overline{PD}$ bisects $\angle QPR$, $\overline{QA} \perp \overline{PD}$, and $\overline{BC}$ passes through A such that $\overline{BC} \parallel \overline{PR}$.

- (a) Prove that $\overline{AC}$ is a median of $\triangle PAQ$.
 (b)★ Find DB . **Hints:** 532, 226, 125



Extra! If at first the idea is not absurd, then there is no hope for it.

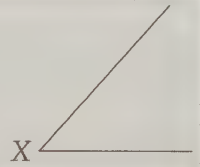
—Albert Einstein

7.7 Construction: Bisectors

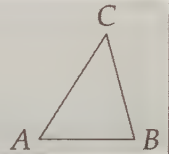
In this section we'll construct some of the special points, lines, and circles of a triangle.

Problems

Problem 7.31: Given $\angle X$ shown, construct a ray from X that bisects $\angle X$.



Problem 7.32: Construct the incircle of $\triangle ABC$ shown.



Problem 7.33: Yes, it's a circle. But you'll note that we haven't marked the center. Your job is to construct it.



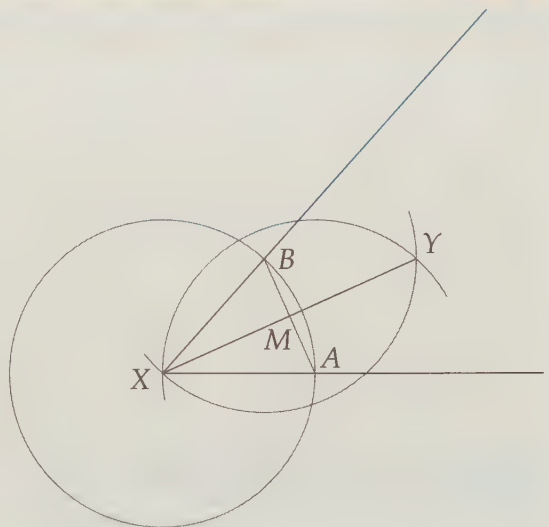
We already know how to make perpendicular lines, midpoints, and perpendicular bisectors. But we haven't learned how to make angle bisectors, so we'll start there.

Problem 7.31: Given $\angle X$, construct a ray from X that bisects $\angle X$.

Solution for Problem 7.31: Since we measure angles as a portion of a circle, we start with a circle centered at X . We call the points where the circle meets the sides of the angle points A and B . Since $XA = XB$, $\triangle XBA$ is isosceles. This makes us happy because we know that the angle bisector of the vertex angle of an isosceles triangle is also a median, an altitude, and a perpendicular bisector.

Therefore, all we have to do is construct the perpendicular bisector of $\overline{AB}$ to have the angle bisector of $\angle X$. We can quickly construct this by drawing arcs centered at A and B with radius AX . These arcs meet at X and a point we'll call Y . $\overrightarrow{XY}$ bisects $\angle AXB$. We'll leave the proof that this construction works as an Exercise. $\square$

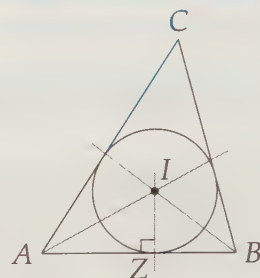
Now that we can construct the special lines of a triangle, we can construct special circles. For example:



Problem 7.32: Given a triangle $\triangle ABC$, construct the incircle of the triangle.

Solution for Problem 7.32: To construct a circle, we need both its center and its radius. We can find the center of the incircle by constructing the bisectors of two of the angles of the triangle. We can perform these constructions exactly as described in Problem 7.31. Where these rays meet is our center, point I .

Now that we have the center, we just need the radius. The radius of the incircle drawn to a point where the incircle touches a side of the triangle is perpendicular to that side. Therefore, we can find the radius by constructing this perpendicular segment from I . Specifically, we construct a line through I that is perpendicular to $\overline{AB}$ as described in Problem 6.24. Suppose this line meets $\overline{AB}$ at Z . IZ is the length of our radius.



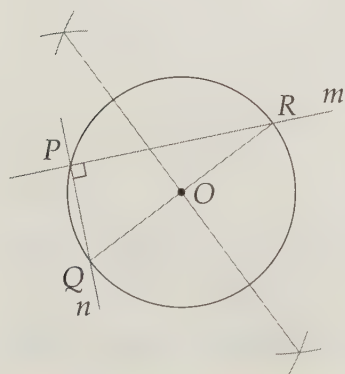
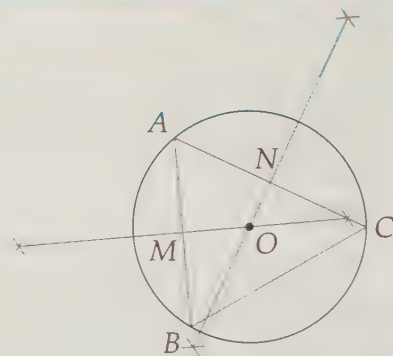
We then construct our incircle by drawing a circle with center I and radius IZ . $\square$

We'll finish this chapter with two clever solutions to a challenging construction problem.

Problem 7.33: Given a circle but not its center, construct the center of the circle.

Solution for Problem 7.33: Solution 1: We don't know a whole lot about constructing parts of a circle, but we know a ton about constructing parts of a triangle. Specifically, we know how to construct the circumcircle of a triangle – its center is just the intersection of the perpendicular bisectors of the sides.

Therefore, we just pick three points A , B , and C on the circumference of our circle. We then construct the perpendicular bisectors of two of the sides of $\triangle ABC$ as shown. The intersection of these perpendicular bisectors, point O , is the circumcenter of $\triangle ABC$. Since our starting circle is the circumcircle of $\triangle ABC$, the circumcenter of $\triangle ABC$ is the center of the starting circle.



Solution 2: Another useful circumcenter fact we know is that the circumcenter of a right triangle is the midpoint of its hypotenuse. Inspired by this observation, we create right $\triangle PQR$ by choosing point P on the circle, drawing line n through P to meet the circle again at Q , then constructing line m through P perpendicular to n . Where line m hits the circle again gives us point R .

Since $\triangle PQR$ is a right triangle, the midpoint of its hypotenuse is the center of its circumcircle. Therefore, we find the center of our original circle by constructing O , the midpoint of $\overline{RQ}$. $\square$

Exercises

7.7.1 Given $\triangle ABC$, construct the circumcircle of the triangle.

7.7.2 Construct a 30° angle.

7.7.3 Construct a 45° angle.

7.7.4 Construct a 120° angle.

7.7.5 Given $\triangle ABC$, construct the orthocenter of the triangle.

7.7.6 Prove that the construction described in the solution to Problem 7.31 really does produce an angle bisector.

7.8 Summary

Definitions:

- The **perpendicular bisector** of a line segment is the line through the midpoint of the segment that is perpendicular to the segment.

Important: The perpendicular bisector of a segment is a straight line consisting of all points that are equidistant from the endpoints of the segment.



- The **angle bisector** of an angle is the line that divides the angle into two equal angles.

Important: The angle bisector of an angle consists of all points that are equidistant from the sides of the angle.

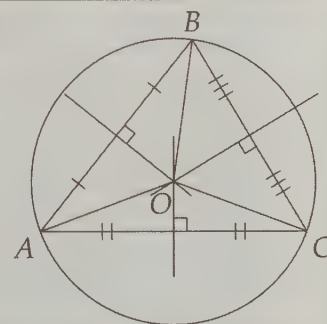


Definition: A **cevian** is a line segment from a vertex of a triangle to a point on the line containing opposite side of the triangle.

We explored important properties of several sets of cevians:

Definitions: The perpendicular bisectors of the sides of a triangle are concurrent at a point called the **circumcenter**. The circle centered at the circumcenter that passes through the vertices of the original triangle is called the **circumcircle** of the triangle because it is **circumscribed** about the triangle (meaning it passes through all the vertices of the triangle).

Finally, the radius of this circle is called the **circumradius**, the circumcenter is usually labeled with the letter O , and the circumradius is usually called R .

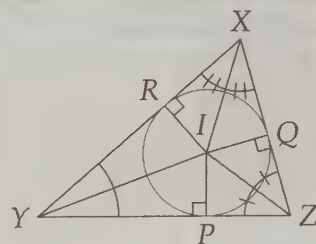


Extra! *There is geometry in the humming of the strings.*

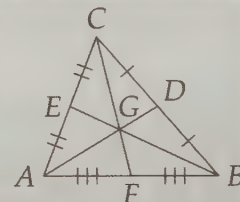


—Pythagoras

Definitions: The angle bisectors of a triangle are concurrent at a point called the **incenter**. This point is equidistant from the sides of the triangle. This common distance from the incenter to the sides of a triangle is called the **inradius**, because the circle with center I and this radius is tangent to all three sides of the triangle. This circle is unsurprisingly called the **incircle** because it is inscribed in the triangle (meaning it is tangent to all the sides of the triangle). The incenter is usually denoted I , and the inradius is usually written as r .



Definitions: A **median** of a triangle connects a vertex of a triangle to the midpoint of the opposite side. The medians of a triangle are concurrent at a point called the **centroid** of the triangle. The centroid of the triangle is usually labeled G .



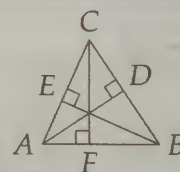
Important:



- The medians of a triangle divide the triangle into six little triangles with equal area.
- The centroid of a triangle cuts its medians into a 2 : 1 ratio. For example, for the triangle shown, we have

$$\frac{AG}{GD} = \frac{BG}{GE} = \frac{CG}{GF} = \frac{2}{1}.$$

Definition: The altitudes of a triangle (extended if necessary) are concurrent at the **orthocenter** of the triangle, which is usually denoted H .



Important:



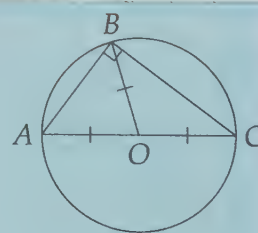
The orthocenter of an acute triangle is inside the triangle. The orthocenter of a right triangle is the vertex of the right angle. The orthocenter of an obtuse triangle is outside the triangle.

Important:



Right triangles offer a few important relationships.

- The circumcenter of a right triangle is the midpoint of the hypotenuse, and the circumradius equals one-half the hypotenuse.
- The median to the hypotenuse of a right triangle is equal in length to half the hypotenuse.



Our exploration of these special points, lines, and circles uncovered a few more important facts:

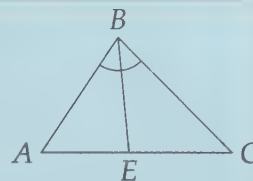
Important: Given any three noncollinear points, there is exactly one circle that passes through all three.



Important: The **Angle Bisector Theorem** states that if E is on $\overline{AC}$ such that $\overline{BE}$ is the angle bisector of $\angle B$ in triangle $\triangle ABC$, then



$$\frac{AB}{AE} = \frac{CB}{CE}.$$



We should think of this any time we have a problem involving lengths and angle bisectors.

Important: The area of a triangle equals its inradius times half its perimeter.



Important: If we connect the midpoints of the sides of a triangle, we divide the triangle into four congruent triangles, each of which is similar to the original triangle. The central triangle is called the **medial triangle** of our original triangle.



Problem Solving Strategies

Concepts:



- If we have the altitudes from two vertices of a triangle, then we know that the line through the intersection point of these two altitudes and the third vertex is also an altitude. Once you have two altitudes, you should almost always think about this third altitude.

This, of course, also works with our other special lines – if you have two medians, you have the third; if you have two angle bisectors, you have the third, etc. With these other special lines you also get a little more, since we have special knowledge about the centroid (the 2 : 1 ratio), the incenter (center of the inscribed circle), and the circumcenter (center of the circumscribed circle).

Again, if you have two of any of these special lines, you should consider the third line and remember what you know about the intersection point.

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Concepts: . . . continued from the previous page



- When given the lengths of the sides of a triangle in a problem, always take the time to check if it is a right triangle. If it is, this fact may simplify the problem considerably.
- Label your diagram with all the lengths and angles you can find. This will help you find relationships in the diagram that might be hard to find otherwise.
- Need to find a length in a problem? Try building right triangles.
- Keep your eye on the ball! Often working backwards from what you want will guide you to the solution.

REVIEW PROBLEMS

7.34 In triangle $\triangle PQR$, $PQ = 12$, $\angle Q = 90^\circ$, and $QR = 16$. Find the area, the circumradius, and the inradius of $\triangle PQR$.

7.35 Points T and U are on $\odot K$ such that K is 2 units from $\overline{TU}$. If $TU = 14$, what is the radius of the circle?

7.36 $\triangle XYZ$ is an equilateral triangle with side length 12. M is the midpoint of side $\overline{YZ}$ and N is the midpoint of $\overline{XZ}$. $\overline{YN}$ and $\overline{XM}$ meet at E .

- What kind of triangle is $\triangle XYM$?
- Find XM .
- Find $[XYZ]$.
- Find XE/EM .
- Find the inradius of $\triangle XYZ$.
- Find the circumradius of $\triangle XYZ$.

7.37 Altitudes $\overline{AD}$ and $\overline{CF}$ of acute triangle $\triangle ABC$ meet at H . Prove that $\angle CHD = \angle ABC$.

7.38 If the altitudes of a triangle all have the same length, must the triangle be equilateral? Why or why not?

7.39 Point D is on side $\overline{YZ}$ of $\triangle XYZ$ such that $\angle YXD = \angle ZXD$. Given that $XZ = 8$, $YD = 3$, and $DZ = 4$, find XY .

7.40 Medians $\overline{AD}$, $\overline{BE}$, and $\overline{CF}$ meet at G . The area of $\triangle ABC$ is 48. Find the areas of $\triangle ADC$, $\triangle AGC$, $\triangle GFB$, $\triangle DEF$, and $\triangle AEF$.

7.41 $\triangle TUV$ is isosceles with $TU = UV = 30$ and $TV = 36$. $\overline{UY}$ bisects $\angle TUV$ and $\overline{TM}$ is a median of $\triangle TUV$. $\overline{UY}$ and $\overline{TM}$ meet at X .

- Show that Y is the midpoint of $\overline{TV}$.
- Show that $\overline{UY} \perp \overline{TV}$.
- Find TY and UY .
- Find XY and XU .
- Find XT and XM .

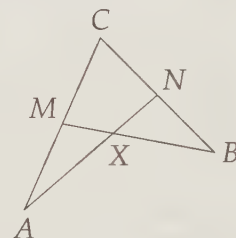
7.42 In Problem 7.8 we found the circumradius of an equilateral triangle with side length 6, while in Exercise 7.2.3, we found the circumradius of an equilateral triangle with side length 18. Given that the first circumradius is $2\sqrt{3}$, how could we have quickly found the second without going through the same laborious procedure we used to solve Problem 7.8?

7.43 In triangle ABC , altitude $\overline{AD}$ intersects angle bisector $\overline{BE}$ at point X . If $\angle BAC = 117^\circ$ and $\angle ACB = 35^\circ$, then determine $\angle DXE$.

7.44 Point O is the circumcenter of $\triangle PQR$, $\angle QPR = 45^\circ$, and $\angle QPO = 23^\circ$.

- Find $\angle RPO$ and $\angle OQP$.
- ★ Find $\angle OQR$.

7.45 M and N are the midpoints of $\overline{CA}$ and $\overline{CB}$, respectively, as shown at right. $\overline{BM}$ and $\overline{AN}$ meet at X . Given that $XM = 3.5$ and $XA = 7.2$, find XN and XB .

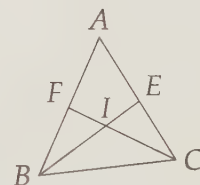


7.46 Medians $\overline{AX}$ and $\overline{BY}$ of triangle $\triangle ABC$ are perpendicular at point O . $AX = 12$ and $BC = 10$.

- Find AO and BY .
- ★ Find the length of median $\overline{CZ}$.

7.47 Cevians $\overline{AD}$ and $\overline{BE}$ meet at point X inside $\triangle ABC$. $\overline{CX}$ extended meets $\overline{AB}$ at F . If $[AEX] = [CEX] = [CDX] = [BDX] = [BXF] = [AXF]$, must X be the centroid of $\triangle ABC$?

7.48 $\overline{BE}$ and $\overline{CF}$ are angle bisectors that meet at I as shown at right. $CE = 4$, $AE = 6$, and $AB = 8$.



- Prove that $\angle EIC = 90^\circ - \frac{\angle A}{2}$.
- Find BC .
- Find BF .

7.49

- Show that if the centroid and the orthocenter of a triangle are the same point, then the triangle is equilateral.
- Show that if the incenter and the centroid of a triangle are the same point, then the triangle is equilateral.

Challenge Problems

7.50 The angle bisector of $\angle G$ of $\triangle GHI$ passes through the circumcenter of $\triangle GHI$. Show that $GH = GI$.
Hints: 454

7.51 Let H and O denote the orthocenter and circumcenter of acute triangle ABC , respectively. $\overleftrightarrow{AH}$ meets $\overleftrightarrow{BC}$ at P and $\overleftrightarrow{AO}$ meets $\overleftrightarrow{BC}$ at Q . Prove that $\angle BAP = \angle CAQ$. **Hints:** 499, 288

7.52 In $\triangle JKL$, we have $JK = JL = 25$ and $KL = 40$. Find the following:

- the area of $\triangle JKL$.
- the inradius of $\triangle JKL$.
- the circumradius of $\triangle JKL$. **Hints:** 305

7.53 Let I be the incenter of triangle ABC , and let $\overleftrightarrow{AI}$ meet $\overleftrightarrow{BC}$ at A' .

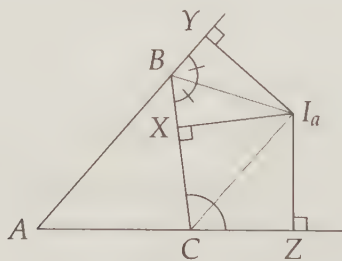
- Prove that $A'I/IA = BA'/AB$.
- ★ Prove that $A'I/IA = BC/(AB + AC)$. **Hints:** 372

7.54 In triangle ABC , $\angle A = 100^\circ$, $\angle B = 50^\circ$, and $\angle C = 30^\circ$. H is on $\overleftrightarrow{BC}$ and M on $\overleftrightarrow{AC}$ such that $\overleftrightarrow{AH}$ is an altitude and $\overleftrightarrow{BM}$ is a median. Find $\angle MHC$. (Source: AHSME) **Hints:** 464

7.55 Is it possible for two angle bisectors in a triangle to be perpendicular? Why or why not? **Hints:** 509

7.56 In $\triangle ABC$, $\angle C$ is a right angle. Point M is the midpoint of $\overleftrightarrow{AB}$, point N is the midpoint of $\overleftrightarrow{AC}$, and point O is the midpoint of $\overleftrightarrow{AM}$. The perimeter of $\triangle ABC$ is 112 and $ON = 12.5$. What is the area of $MNCB$? (Source: MATHCOUNTS)

7.57 For a given triangle ABC , extend $\overleftrightarrow{AB}$ and $\overleftrightarrow{AC}$ as shown below. We will prove that there is a circle tangent to $\overleftrightarrow{BC}$ and the extensions of $\overleftrightarrow{AB}$ and $\overleftrightarrow{AC}$. We start by drawing the bisectors of two of the exterior angles of $\triangle ABC$ as shown with the grey lines below.



- Let P be a point on the external angle bisector of angle $\angle B$. Prove that P is equidistant from $\overleftrightarrow{AB}$ and $\overleftrightarrow{BC}$.
- Let Q be a point on the external angle bisector of angle $\angle C$. Prove that Q is equidistant from $\overleftrightarrow{AC}$ and $\overleftrightarrow{BC}$.

- (c) Let I_a be the intersection of the external angle bisectors of angles $\angle B$ and $\angle C$, and let X , Y , and Z be the feet of the altitudes from I_a to $\overline{BC}$, $\overline{AB}$, and $\overline{AC}$. Prove that $I_aX = I_aY = I_aZ$. What does this tell us about the circle with center I_a and radius I_aX ? This circle is called an **excircle** of $\triangle ABC$; I_a is called an **excenter**, and r_a is an **exradius**. Every triangle has three excircles, one tangent to each side of the triangle.
- (d) Prove that I_a lies on the internal angle bisector of angle $\angle A$.
- (e)★ Let $r_a = I_aX$. Prove that $r_a = \frac{[ABC]}{s - BC}$, where s is the semiperimeter of $\triangle ABC$. **Hints:** 44, 474

7.58 Points M and N are the midpoints of sides $\overline{PA}$ and $\overline{PB}$ of $\triangle PAB$. As P moves along a line that is parallel to side $\overline{AB}$, which of the four quantities listed below change? (Source: AMC 10)

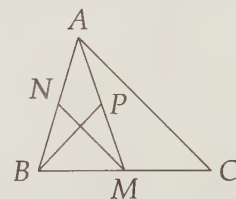
- (a) the length of segment $\overline{MN}$.
- (b) the perimeter of $\triangle PAB$.
- (c) the area of $\triangle PAB$.
- (d) the area of $ABNM$.

7.59 Let $\triangle XYZ$ have $\angle X = 60^\circ$ and $\angle Y = 45^\circ$. A circle with center P passes through A and B on side $\overline{XY}$, C and D on side $\overline{YZ}$, and E and F on side $\overline{ZX}$. Suppose $AB = CD = EF$.

- (a) Prove that P is the incenter of $\triangle XYZ$.
- (b) Find $\angle XPY$. (Source: HMMT)

7.60★ A ladder is initially resting vertically against a wall that is perpendicular to the ground. It begins to slip and fall to the ground, with the bottom of the ladder moving directly from the wall, and the top of the ladder always touching the wall. What path does the midpoint of the ladder trace? (Don't forget to prove that the midpoint of the ladder really does hit every point on the path!) **Hints:** 271, 347

7.61 Triangle $\triangle ABC$ has area 48. Let P be the midpoint of median $\overline{AM}$ and let N be the midpoint of side $\overline{AB}$. If G is the intersection of $\overline{MN}$ and $\overline{BP}$, find the area of $\triangle MGP$. (Source: Mandelbrot) **Hints:** 212, 407



7.62★ In this problem we will prove that the medians of a triangle are concurrent. Let $\triangle ABC$ be our triangle. Medians $\overline{AD}$ and $\overline{BE}$ meet at point G . $\overline{CG}$ meets $\overline{AB}$ at point F .

- (a) Show that $[ACG] = [GCB] = [AGB]$. **Hints:** 557
- (b) Use your result from part (a) to show that $AF = FB$.
- (c) Do the previous parts show that the medians of any triangle are concurrent?

7.63★ Point N is on hypotenuse $\overline{BC}$ of $\triangle ABC$ such that $\angle CAN = 45^\circ$. Given $AC = 8$ and $AB = 6$, find AN . **Hints:** 448, 560, 585

7.64★ The circle at right has radius 1 and is circumscribed about equilateral triangle ABC . If X is the midpoint of $\overline{AC}$ and Y is on arc $\widehat{AC}$ such that $\angle YXA$ is right, then find length XY . (Source: Mandelbrot)

